

Logit Prisms: Decomposing Transformer Outputs for Mechanistic Interpretability

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Abstract

We introduce a straightforward yet effective method to break down transformer outputs into individual components. By treating the model’s non-linear activations as constants, we can decompose the output in a linear fashion, expressing it as a sum of contributions. These contributions can be easily calculated using linear projections. We call this approach “logit prisms” and apply it to analyze the residual streams, attention layers, and MLP layers within transformer models. Through two illustrative examples, we demonstrate how these prisms provide valuable insights into the inner workings of the `gemma-2b` model.

1 Introduction

The logit lens (nostalgebraist, 2020) is a simple yet powerful tool for understanding how transformer models (Brown et al., 2020; Vaswani et al., 2017) make decisions. In this work, we extend the logit lens approach in a mathematically rigorous and effective way. By treating certain parts of the network activations as constants, we can leverage the linear properties within the network to break down the logit output into individual component contributions. Using this principle, we introduce simple “prisms” for the residual stream, attention layers, and MLP layers. These prisms allow us to calculate how much each component contributes to the final logit output.

Our approach can be thought of as applying a series of prisms to the transformer network. Each prism in the sequence splits the logits from the previous prism into separate components. This enables us to see how different parts of the model—such as attention heads, MLP neurons, or input embeddings—influence the final output.

To showcase the power of our method, we present two illustrative examples:

- In the first example, we examine how the `gemma-2b` model performs the simple factual retrieval task of retrieving a capital city from a country name. Our findings suggest that the model learns to encode information

about country names and their capital cities in a way that allows the network to easily convert country embeddings into capital city unembeddings through a linear projection.

- The second example explores how the gemma-2b model adds two small numbers (ranging from 1 to 9). We uncover interesting insights into the workings of MLP layers. The network predicts output numbers using interpretable templates learned by MLP neurons. When multiple neurons are activated simultaneously, their predictions interfere with each other, ultimately producing a final prediction that peaks at the correct number.

2 Method

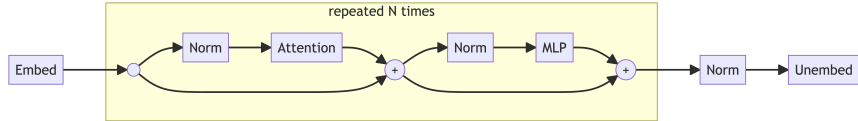


Figure 1: A typical decoder-only transformer network where the residual stream is iteratively refined by a sequence of attention and MLP layers. There are five main components: the input embedding (Embed), the normalization layers (Norm), the Attention layers, the MLP layers, and the unembedding layer (Unembed).

We introduce simple “prisms” that allow us to break down the output of transformer networks into individual components. The key idea is to treat nonlinear activations as constants, which enables us to calculate the contribution of any component in the network using a series of linear transformations. In the following subsections, we explore this approach in more detail.

2.1 Residual Stream Decomposition

In a typical decoder-only transformer architecture (see Figure 1), the output logit can be expressed as:

$$\text{logits} = W_{\text{unembed}} \cdot \text{diag}(w_{\text{norm}}) \cdot \frac{w_{\text{embed}} + \sum_{i=1}^N (a_i + m_i)}{s}$$

Here, $W_{\text{unembed}} \in \mathbb{R}^{V \times d}$ is the unembedding matrix, w_{embed} is the token embedding vector, a_i and m_i denote the attention and MLP outputs at i -th layer respectively, s is a normalization factor (usually the root mean square of the denominator), and w_{norm} is the scaling vector of the last normalization layer.

By treating s as a constant, we can break down the logits into individual terms for each embed, attention and MLP layer:

$$\text{logits} = P \cdot w_{\text{embed}} + P \cdot a_1 + P \cdot m_1 + \dots + P \cdot a_N + P \cdot m_N$$

where P is the projection matrix:

$$P = W_{\text{unembed}} \cdot \text{diag}(w_{\text{norm}}) \cdot (s^{-1}I)$$

Each term, like $P \cdot a_1$, represents the individual contribution of the corresponding residual vector (a_1) to the final logit score. The cumulative sum $P \cdot w_{\text{embed}} + \sum_{i=1}^{\ell} P \cdot (a_i + m_i)$ gives the model's logit score up to layer ℓ .

2.2 Attention Decomposition

We can naturally break down the attention output into a sum over its attention heads:

$$a_{\ell} = \sum_{i=1}^{\text{\#heads}} H_{\ell}^i$$

Here, H_{ℓ}^i is the i -th attention head's output at layer ℓ . Additionally, an attention head's output is a weighted sum of the OV circuit outputs for each token in the sequence.¹ The attention output at layer ℓ is:

$$H_{\ell}^i = \sum_p \alpha_p \cdot OV_{\ell}^i \cdot \text{diag}(w_{\text{norm}}) \cdot \frac{h_{\ell-1}^p}{s_{\ell-1}^p}$$

In this formula, α_p is the attention weight from the current token to the previous token p . OV_{ℓ}^i is the OV matrix for the i -th attention head at layer ℓ . $h_{\ell-1}^p$ is the hidden state of token p from the previous layer $\ell - 1$. Finally, $s_{\ell-1}^p$ represents the normalization factor.

By treating the attention weight α_p and normalization factor $s_{\ell-1}^p$ as constants, we can express the attention output as a sum of linear projections of the previous layer's hidden state $h_{\ell-1}^p$. To find the contribution of token p via attention head i , we use the following projection matrix:

$$P_{\ell}^{a_i} = P \cdot (\alpha_p I) \cdot OV_{\ell}^i \cdot \text{diag}(w_{\text{norm}}) \cdot (s_{\ell-1}^p)^{-1} I$$

Note that we can further decompose $h_{\ell-1}^p$ into the sum of the token embedding and all previous layers' residual outputs using residual stream decomposition.

2.3 MLP Decomposition

In transformer networks, the MLP layers consist of two linear transformations, W_{up} and W_{down} , with a non-linear function g applied between them. The output of an MLP layer ℓ can be expressed as:

$$m_{\ell} = W_{\text{down}} \cdot g \cdot W_{\text{up}} \cdot \text{diag}(w_{\text{norm}}) \cdot \frac{h_{\ell}^a}{s_{\ell}^a}$$

¹The OV circuit has two matrices: V reads from the residual stream, and O writes to it.